The Perfect Answer Revision Guide CIE Igcse Biology 2

SCIENCE with HAZEL

The Perfect Answer Revision Guide To...

Biology

CIE IGCSE

9-1 / A*-U

1st Edition

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NOTE: Core content is given in this format (Grades 1-5). *Supplementary content is given in italics (grades 5-9)*

1. **Characteristics and classification of living organisms**

Characteristics all living organisms show:
- Movement
- Respiration
- Sensitivity
- Nutrition
- Excretion
- Reproduction
- Growth

Define movement
- Action by all *or part* of an organism
- Causes change in position or place

Define respiration
- Chemical reactions in cells
- Nutrient molecules broken down
- Energy released *for metabolism*

Define sensitivity
- Detecting and responding to changes in the environment
- *Sensing stimuli in the internal or external environment*
- *Making appropriate responses*

Define growth
- Permanent increase in size
- *Increase in dry mass*
- *Cell numbers and/or cell size increase*

Define reproduction
- Processes that make more of the same kind of organism

Define excretion
- Removal from organisms of toxic materials and substances in excess of requirements
- *Removal of waste products of metabolism*

Define Metabolism
- *Chemical reactions in cells including respiration*

Define nutrition
- Taking in materials for energy, growth and development

Plants require:
- *Light*
- *Carbon dioxide*
- *Water*
- *Ions*

Animals require:
- *Organic compounds*
- *Ions*
- *Usually need water*

What is a classification system?

- System to classify organisms in to groups by the features they share

What is a *species?*

- Group of organisms that can reproduce to make fertile offspring

The *binomial* system

- Internationally agreed system to classify organisms
- Two-part scientific names showing genus and species (e.g. *Canis lupus*)

Traditional classification systems

- *Reflect evolutionary relationships*
- *Based on morphology and anatomy*

What is a more accurate way to classify organisms?

- *Sequence of bases in their DNA*
- *Sequences of amino acids in their proteins*

Explain how DNA sequence and ancestry are related

- *More recent ancestors share more similar base sequences in DNA*

Cells of all living organisms:

- Cytoplasm
- Cell membrane
- DNA (as genetic material)
- *Ribosomes for protein synthesis*
- *Enzymes involved in respiration*

List the 5 Kingdoms:

- Animals
- Plants
- Fungi
- Prokaryotes
- Protoctists

What feature makes a prokaryote?

- *No definite nucleus*

What feature makes a protoctist?

- *All the cells are the same*

Plants:

- Have different types of cells
- Have cell walls

Fungi:

- Have cell walls
- Feed by external digestion

Animals:

- No cell wall
- Different types of cells
- Internal digestion

What is a vertebrate?

- An animal with a backbone

The main groups of vertebrates

Mammals
- Furry skin
- Mammary glands

Birds
- Wings
- Feathers
- Scales on legs and feet

Reptiles
- Dry scaly skin

Fish
- Have scales and fins
- Operculum covering gills
- Lateral line containing sense organs

Amphibians
- Moist skin

The main groups of arthropods

Myriapods
- Antennae
- Many body segments
- Hard exoskeleton

Insects
- Compound eyes
- Segmented body- head, thorax and abdomen
- Two pairs of wings
- Antennae
- Three pairs of legs
- Mouthparts

Arachnids
- Head and thorax combined
- Simple eyes
- Four pairs of legs
- Spinneret
- Powerful jaws (all predators)

Crustaceans
- Claws with hard serrated edges
- Eyes
- Jointed limbs
- Gills under shell
- Carapace

Features of a fern
- *Leaves called fronds*
- *Fronds carry sporangia*
- *Sporangia release spores*
- *Have underground rhizomes*
- *Simple, true roots*

Features of an angiosperm

- *Produce flowers*
- *Produce fruit*
- *Can be monocotyledons or dicotyledons*
- *Have stomata*
- *Vascular systems for transport*
- *Extensive root systems*

What is a virus made up of?

- *Protein coat*
- *Genetic material*

2. Organisation of the organism

Plant and animal cells both have:

- Nucleus
  - Contains genetic material (DNA) - controls activities and characteristics of cell
  - Chromosomes visible during cell division
- Cytoplasm
  - Contains water and dissolved substances
  - *Contains mitochondria for aerobic respiration*
  - *Rough endoplasmic reticulum (rER) to make proteins*
- Cell membrane
  - Surround cytoplasm
  - Controls entry and exit of dissolved substances
  - Forms barrier between cell and surroundings

Plants also have:

- Chloroplasts
  - Packed with chlorophyll
  - Contain enzymes to produce glucose through photosynthesis
- Vacuole
  - Contains water
  - Provides turgor pressure to maintain shape of cell
  - Filled with cell sap
- Cell wall
  - Protects and supports the cell

Which organisms do not have mitochondria and rER?

- *Prokaryotes*

What adaptation do high-energy cells (such as muscle) have?

- *Need more energy, so have more mitochondria*

Specialised cells

Describe the structure and function of the following specialised cells…

Animal cells

Red blood cell

- Function: transport oxygen around the body for respiration
- Contains haemoglobin which binds to oxygen forming oxyhaemoglobin
- Biconcave disc shape increases surface area to volume ratio
- Very flexible enabling it to pass through small vessels
- No nucleus - more room for haemoglobin so more oxygen can then be transported

Muscle cell

- Function: to contract and relax in order to move muscles
- Cells are long
- Contain many protein fibres
- Protein fibres can shorten the cell when energy is available
- This makes the cell able to contract

Ciliated cell

- Function: move mucus out of the trachea and bronchi
- Have layer of tiny hairs (cilia)
- Cilia can move and push mucus
- Mucus traps dust and microbes
- Mucus is expelled

Motor nerve cell
- Function: conduct impulses
- Long fibre called an axon
- Axon can carry electrical impulses
- Axons have a fatty sheath to insulate
- Many branched ending can connect with other cells

Sperm cell
- Function: reproduction
- Haploid nucleus
- Flagellum beats to swim cell towards ovum
- Acrosome contains enzymes to penetrate egg cell
- Mitochondria in cytoplasm release energy for movement

Egg cell (ovum)
- Function: reproduction
- Haploid nucleus
- Jelly coat changes at fertilisation
- Allows entry of male nucleus

Plant cells

Root hair cell
- Function: absorption
- Long extension called root hair
- Large surface area
- For absorption of water by osmosis
- For absorption of mineral ions by active transport

Xylem vessel
- Function: transport and support
- No end wall, so many cells can form a tube
- No cytoplasm, so water can pass freely
- Coated with lignin to strengthen and support plant
- Transports mineral ions and water from roots upwards

Palisade mesophyll
- Function: photosynthesis
- Contains many chloroplasts to absorb light
- Tall thin cells - densely packed
- Maximises photosynthesis

Define tissue
- Group of cells with similar structures working together to perform the same function

Define organ
- Group of tissues
- Work together to perform specific functions

Define organ system
- Group of organs with related functions
- Work together to perform bodily functions

e.g. Digestive system:

- Tissues - glandular and muscular tissue
- Organs - stomach, small intestine, oesophagus, liver, large intestine
- Organ system - digestive system

Size of specimens

How do you calculate magnification?

- Magnification = measured length/ actual length

How do you calculate Actual (true) length?

- Actual length = Measured length / magnification

3. Movement in and out of cells

What is diffusion?

- Net movement of particles
- From region of their higher concentration to lower concentration
- Due to random movement

Where does the energy for diffusion come from?

- *The kinetic energy from molecules and ions*

Why is diffusion important?

- Many life processes depend on it:
    - Oxygen entering blood from lungs
    - Glucose and amino acids pass from gut to blood
    - Plants absorb CO_2 in to leaves

How do substances enter cells?

- By diffusion through the cell membrane

How can diffusion speed be increased in some organisms?

- *Reduced diffusion distances: Shorter distance = faster diffusion*
- *Concentration gradients kept high: Equilibrium is not reached so diffusion continues*
- *Large surface area for diffusion to take place across* cell membranes

What is osmosis?

- Diffusion of water through partially permeable membranes
- *Net movement of water molecules from region of high water potential to low water potential*
- *Through a partially permeable membrane*

How does water enter cells?

- By osmosis through the cell membrane

How does osmosis affect plant and animal cells differently?

- *Plant cells not permanently damaged*
- *Cell walls give plant cell support*
- *Animal cells permanently damaged*
- *Animal cells shrink with water loss and burst if swell too much*

What is plasmolysis?

- *When too much water moves out of a plant cell the cell contents shrink*
- *This pulls the cell membrane away from the cell wall*

What is the result of putting plant tissue in high concentration solution?

- Plant cells have higher water potential
- Water lost from plant into solution
- *Plant cells become flaccid*

If plant tissue placed in dilute solution

- Plant cells have lower water potential
- Water enters plant through osmosis
- *Plant cells become more turgid*

Plants are supported by

- Water pressure inside cells pushing outwards
- *Water pressure acts against inelastic cell wall - turgor pressure*

Define active transport

- Movement of particles through a cell membrane
- From lower to higher concentration
- Requires energy from respiration

Where is active transport useful?

- *Can move molecules up a concentration gradient*
- *E.g. Uptake of ions in to root cells, uptake of glucose in kidney tubules*

Describe how active transport occurs

- *Protein carriers are found in the cell membrane*
- *Carrier recognises particle to transport*
- *Particle transported across membrane by carrier protein using energy*
- *Particle released in to cell*
- *Carrier protein returns to surface membrane*

4. Biological molecules

What are the chemical components of organic molecules (including carbohydrates and lipids)?
- Carbon
- Hydrogen
- Oxygen
- Proteins also contain Nitrogen and Sulphur
- Nucleic acids also contain Phosphorous and Nitrogen

What is a carbohydrate?
- Made of simple sugar molecules
- Monosaccharides e.g. Glucose – one sugar
- Polysaccharides e.g. starch, glycogen, cellulose- many sugars

What is a lipid?
- Made up of fatty acids and glycerol
- 3 fatty acids to 1 glycerol molecule
- Fats are solid at room temp, oils are liquid

What is a protein?
- Chains of amino acids
- Sequence is coded for by genes
- *Sequence of amino acids determines protein shape*
- *Protein shape and structure gives the protein its function*

How does the shape of an enzyme determine its function?
- *Enzyme has active site*
- *Active site has specific shape*
- *Only target molecule can fit in the active site*

How does the structure of an antibody allow it to work?
- *Antibodies have a binding site*
- *Binding site has specific shape*
- *Can therefore recognise specific antigens*

What is the food test for starch?
- Iodine
- Positive result: turns blue/black

What is the test for glucose?
- Heat with water and Benedict's reagent
- Blue solution turns brick red in the presence of glucose

What is the test for protein?
- Biuret reagent
- Positive result: purple

What is the test for fat?
- Add ethanol
- Add water
- Positive result: milky white emulsion

What is the test for Vitamin C?
- DCPIP dye
- Add vitamin C drop by drop
- Few drops= strong solution

Describe the structure of DNA

- *Two strands coiled in a double helix*
- *Strands contain bases*
- *Cross links formed by base pairs*
- *Adenine (A) with Thymine (T)*
- *Cytosine (C) with Guanine (G)*

What is the role of water in organisms?

- Important as a solvent *for digestion, excretion and transport*

5. Enzymes

What is a catalyst?
- Increases rate of chemical reaction
- Not changed or used up by the reaction

Define enzyme
- Protein that functions as a catalyst

Why are enzymes important?
- Speed up biochemical reactions
- High reaction speed necessary to sustain life

How does an enzyme work?
- Enzyme has a complementary shape to substrate
- Substrate molecules bind
- *Enzyme-substrate complex formed at active site*
- *Complementary shape of active allows specificity*
- Product is produced
- *Each enzyme is specific for one shape of substrate*
- *One reaction occurs each time substrate binds*

How does increasing temperature affect enzymes?
- Up to optimum temperature: enzyme activity increases with increasing temperature
- *Enzyme and substrate have more kinetic energy*
- *More likely to bind to active site due to more effective collisions*
- Over optimum temperature: enzyme activity decreases with increasing temperature
- *Over optimum temperature enzyme loses shape - denatured*
- *Denatured enzyme cannot bind substrate*

How does change in pH affect enzymes?
- Activity highest at optimum pH
- Above or below optimum, activity decreases
- *Changes in pH alter 3D shape*
- *If 3D shape changes, enzyme is denatured and cannot fit substrate*

6. Plant nutrition

What is photosynthesis?

- Process plants use to produce carbohydrates
- Uses energy from light

What is the word equation for photosynthesis?

- Carbon dioxide + Water —> Glucose + Oxygen
- Requires light and chlorophyll
- *$6CO_2 + 6H_20$ —> $C_6H_{12}O_6 + 6O_2$*

What is the function of chlorophyll?

- *Transfers light energy to chemical energy*
- *Energy produced used for synthesis of carbohydrates*

Limiting factor

- *Something in such short supply it restricts life processes*
- *E.g. light, carbon dioxide, water*

What factors can limit photosynthesis?

- *Sunlight in cloudy/dark places*
- *Water in dry places*
- *Temperature in too hot or too cold conditions*
- *Carbon dioxide*

How can photosynthesis be optimised in temperate or tropical countries?

- *Use a glasshouse to control variables*
- *In a cold country, increase temperature to optimum level*
- *In a dry environment, increase humidity/water available*
- *Ensure optimum light using electric lighting*
- *Choose plant species which suit the environment*
- *Use automatic systems to control variables*

How is the internal structure of the leaf adapted for photosynthesis?

- *Palisade mesophyll: tightly packed to maximise light absorption, contain lots of chloroplasts*
- *Chloroplasts: can move to maximise absorption of light*
- *Waxy cuticle / Upper epidermis: transparent to allow light to reach chloroplasts, waxy to prevent evaporation of water and stops pathogen entry*
- *Air spaces in spongy mesophyll - allow gases to diffuse*
- *Xylem: allows entry of water and mineral ions by transpiration stream*
- *Guard cells: control opening and closure of stomata*
- *Stomata: allow carbon dioxide to enter, oxygen and water to leave*
- *Leaf shape: thin and broad (large surface area)*

What is the importance of nitrate ions for amino acids?

- Amino acids contain nitrogen
- Amino acids form proteins
- *Deficiency causes stunting, weakness, yellowing*

How are magnesium ions important?

- Magnesium forms part of chlorophyll
- *Deficiency causes chlorosis - leaves turn yellow*

7. Human nutrition

What is a balanced diet?
- Provides all the nutrients in correct amounts needed to carry out life processes

How does age affect dietary needs?
- Children need less total dietary energy (kJ) than adults

How does gender affect dietary needs?
- Men need more total energy (kJ) than women

How do activity levels affect dietary needs?
- Increased activity levels need more total energy (kJ)

How does pregnancy affect dietary needs?
- Pregnant women need more total energy (kJ) than non-pregnant women

What is the energy requirement in pregnant and breastfeeding mothers?
- Pregnancy needs less energy than breastfeeding

What is deficiency in total energy AND protein called?
- Marasmus

Nutrient	Dietary importance	Principal source	Deficiency Disease
Protein	Growth and repair of muscles, enzymes, hormones, antibodies	Meat, fish, eggs, legumes, mycoprotein	Kwashiorkor
Carbohydrates	Energy store	Rice, Potatoes, wheat, cereals	
Fat	Insulation and concentrated energy store	Meat, eggs, milk, cheese	
Water	70% of body. Tissue fluid, cytoplasm, blood.		Dehydration
Fibre (roughage)	prevents constipation, encourages peristalsis		Constipation
Calcium	Strong teeth and bones	Milk, cheese and fish	weak bones, poor clotting, spasms, rickets
Iron	Healthy blood	Red meat, liver, leafy greens	Anaemia
Vitamin C	Sticks together cells lining the mouth	Citrus fruits, leafy greens	Scurvy - bleeding gums
Vitamin D	Strong bones	Liver, dairy products, eggs	Rickets

The digestive process

Define ingestion
- When food/drink enters the mouth

Define mechanical digestion
- Breakdown of food to smaller pieces, no chemical change
- E.g. chewing, stomach muscles churning food

Define chemical digestion
- Breakdown of large insoluble molecules to small soluble ones
- E.g. amylase in saliva breaks down starch into simple sugars

Define absorption
- Movement of small food molecules and ions through intestine wall into blood stream

Define assimilation
- Movement of digested food into cells where they are used

Define excretion
- The removal of waste products of metabolism
- E.g. sweat, carbon dioxide

Define egestion
- The removal of food not digested or absorbed (faeces) from the anus

What is diarrhoea?
- Loss of watery faeces

How is diarrhoea treated?
- Oral rehydration therapy - mixture of water and mineral salts
- Prevents dehydration

What is cholera?
- Bacterial disease
- *Bacteria produces toxin*
- *Toxin increases loss of chloride ions in small intestine*
- *Leads to osmotic loss of water into gut*
- *Causes diarrhoea, dehydration, loss of salts from blood*

Site	Function
Mouth	Food converted to bolus by teeth during mastication (chewing)
Salivary Glands	Produce saliva, transport to mouth by salivary ducts
Oesophagus	Food moves to stomach by peristalsis (contraction of circular muscles)
Stomach	Food mixes with hydrochloric acid - forms chyme
Small Intestine (Duodenum And Ileum)	Covered in villi to to increase absorption of digested food
Duodenum	Semi liquid food mixes with pancreatic juice
Ileum	Digested food absorbed in to blood

Liver	Produces bile - neutralizes acid and emulsifies fat in the small intestine
Pancreas	Produces pancreatic juice, transported into small intestine by pancreatic duct
Gall Bladder	Stores bile - transported in to duodenum by bile duct
Large Intestine - Colon	Reabsorbs water
Rectum	Stores faeces
Anus	Exit for faeces- two sphincter muscles

What is the function of the teeth?
- Mechanical digestion
- Incisors cut/bite
- Canines hold/cut
- Premolars crush and chew
- Molars grind and chew

Describe the structure of human teeth

Structure	Function
Root	Embedded in the gum to anchor the tooth in the mouth
Enamel	Hardest substances made by animals. Covers the tooth and provides a tough surface for biting and chewing
Dentine	Bone-like structure under the enamel. Contains cytoplasm and tubes running from the pulp cavity outwards which are filled with blood vessels and nerves
Pulp Cavity	Hollow middle of the tooth. Contains nerves and blood vessels which supply the cytoplasm with food and oxygen
Cement	Covers the root of the tooth
Fibres	Grow out of the cement and attach the tooth to the jawbone
Nerves	Allow teeth to sense pressure and pain

What are the causes of tooth decay?
- Bacteria in the mouth stick to teeth and makes plaque
- Bacteria respire sugars, makes acid
- Acid erodes enamel and dentine

Describe proper tooth care
- Diet low in sugar
- Regular effective brushing
- Eat crisp veg or rinse mouth after meals

What is the role of chemical digestion?
- Producing soluble small molecules that can be absorbed and assimilated

What is role of amylase?
- *Secreted into alimentary canal*
- To break down starch into simpler sugars (*maltose)*

Describe the action of maltase
- *Breaks down maltose in to glucose*
- Occurs o*n the membranes of small intestine epithelium*

Where is amylase secreted?
- Mouth and small intestine (salivary and pancreatic)

Where is protease secreted?
- Stomach
- Small intestine

Where is lipase secreted?
- In bile in to small intestine

What is the role of lipase?
- To digest fats into fatty acids and glycerol

What is the role of protease?
- To digest proteins into amino acids

What are pepsin and trypsin?
- *Protease enzymes*
- *Pepsin - works in the stomach*
- *Trypsin - works in the small intestine*

Describe the functions of hydrochloric acid in the stomach
- Killing bacteria in food
- *Low pH denatures enzymes in harmful microorganisms*
- Provides acidic pH for enzymes
- *Optimum pH for pepsin activity*

What is the role of bile?
- *Neutralises acidic food mixture from stomach*
- *Provides suitable pH for enzymes in small intestine*
- *Emulsifies fats to increase surface area for chemical digestion*

Where is digested food absorbed?
- Small intestine

What is the purpose of villi and microviili?
- *Increases surface area for absorption*

Describe the structure of a microvillus
- *Epithelium one cell thick - short diffusion pathway*
- *Goblet cell produces mucus to protect lining of small intestine*
- *Good blood supply from capillaries to transport glucose to rest of body*
- *Lacteal transports fatty acids and glycerol*

Where is water absorbed?
- Small intestine (mostly) and colon

8. Transport in plants

What is the function of the xylem?
- Water carried from roots to leaves in xylem
- Transports water and minerals **up** the plant

What is the function of the phloem?
- Carries sugar from the leaves **up and down** rest of plant

How is a root hair cell adapted?
- *Large surface area*
    - o *Increases rate of water absorption by osmosis*
    - o *Increases ion uptake by active transport*

Describe the pathway of water in a plant
- Root hair cell
- Root cortex
- Xylem
- Mesophyll cells

How could you investigate the path of water in a plant?
- Cut a piece of celery
- Put in water
- Add stain (methylene blue or eosin)
- Wait a few hours and cut surface - dye appears in xylem

What is transpiration?
- Loss of water vapour from leaves by evaporation at surface of mesophyll cells
- Water vapour then diffuses through stomata
- *Water loss related to large surface area of leaf*
- *Water evaporates in to air spaces in plant*
- *Water diffuses out of stomata down concentration gradient*
- *Loss of water at leaves reduces water potential*
- *Causes transpiration pull*
- *Column of water drawn up xylem, held by cohesion*

What is wilting?
- *Water loss exceeds uptake, cells go flaccid*

Investigate effect of temperature and humidity on transpiration rate
- Use bubble podometer
- Connect plant to podometer with rubber tubing
- Record speed of air bubble in capillary tube (dependent variable)
- Vary temp or humidity as independent variables
- Increased humidity slows transpiration —> *Decreases water potential gradient*
- Increased temperature increase transpiration —> *Increases water holding capacity of air*

Describe and explain how rates of transpiration may be increased
- *Hot - water evaporates faster*
- *Dry - increases the concentration gradient between the leaf and the surrounding air*
- *Windy - water is blown off the leaf increasing concentration gradient*

Describe and explain how rates of transpiration may be decreased
- *Cold - water evaporates slower*
- *Humid - small concentration gradient between the leaf and the surrounding air*
- *Still air - water is not removed from the surface of the leaf*

What is translocation?

- *Movement of sucrose and amino acids*
- *From source (where produced) to regions of storage OR where used in respiration (sink)*
- *Some parts of plant act as source and sink at different times*

9. Transport in animals

What is the circulatory system?

- System of blood vessels with pump and valves causing one way movement of blood

Describe circulation in a fish

- *Single circulation, blood flows through heart once per circuit*
- *Blood pressure lower than mammals - too low for kidney function*

What are the advantages of double circulation?

- *Blood passes through heart twice per circuit*
- *Increased blood pressure through lungs*
- *Increased pressure to tissues*

Explain the thickness of the heart walls

- *Left ventricle thicker - must push blood round whole body*
- *Right wall thinner as only pushes blood to lungs*
- *Atria walls thinner than ventricles, only push to ventricles*

What is the function of the septum?

- *Separates oxygenated and deoxygenated blood in the heart*

Describe the movement of the blood around the body starting at the right atrium

- *Deoxygenated blood enters the right atrium via the vena cava*
- *Right atrium contracts forcing blood through tricuspid valve into right ventricle*
- *Blood enters the right ventricle and leaves via the pulmonary artery*
- *Blood flows to the lungs*
- *Blood become oxygenated*
- *Blood returns to the heart and enters the left atrium via the pulmonary vein*
- *Left atrium contracts forcing blood through bicuspid valve into the left ventricle*
- *Left ventricles contracts forcing blood into the aorta*
- *Oxygenated flows around the body and is used in respiration*

How can heart activity be monitored?

- By an ECG, pulse rate and listening to valves with stethoscope

What does physical activity do to pulse rate?

- Increases it

How would you investigate effect of exercise on pulse rate?

- Take pulse at rest, perform exercise, take pulse rate again

What is coronary heart disease?

- Blockage of coronary arteries
- Part of heart muscle stops contracting, causing heart attack
- Vital tissues don't get oxygen

What are the risk factors?

- Diet
- Stress
- Smoking
- Genetic predisposition
- Age
- Gender

Explain the effect of exercise on pulse rate

- *Muscles burn more oxygen in respiration*
- *Muscle creates waste products*
- *Increased oxygen demand and waste products needs increased circulation*
- *Pulse rate increases to keep up with demand*

How can diet and exercise prevent coronary heart disease?

- *Poor diet and lifestyle are risk factors*
- *Low levels of cholesterol and low saturated fatty acids in diet*
- *Exercise strengthens heart muscle and reduces stress*

How can heart disease be treated?

- *Drug treatment with e.g. aspirin*
- *Surgery e.g. Bypass operation or angioplasty (stent)*

Blood and lymphatic vessels

Vessel	Description	*How is it adapted to function?*
Artery	Thick wall, muscular, carries blood away from heart	*Elastic walls expand and relax, hick walls withstand high pressure*
Vein	Thin walled, wide lumen, takes blood back to heart	*Valves prevent back flow, low pressure, large diameter creates low resistance*
Capillary	One cell thick, branched structure, extends through all tissues	*Short diffusion distance, large surface area maximises exchange of substances*

Name the main blood vessels going to and from the heart

- Vena cava
- Aorta
- Pulmonary vein

Name the main blood vessels going to and from the lungs

- Pulmonary artery
- Pulmonary vein

Name the main blood vessels going to and from the kidney

- Renal artery
- Renal vein

What is the function of arterioles?

- *Transport blood from arteries to capillaries*
- *Regulate blood flow and pressure*

What is the function of venules?

- *Transport blood from capillaries to veins*

What is the function of shunt vessels?

- *Transports blood directly from artery to vein*
- *Regulates blood flow*

Describe the lymphatic system

- *Network of vessels and nodes*
- *Vessels carry lymph fluid*
- *Lymph nodes have lymphocytes to remove microbes*
- *Blockage can produce severe swelling*
- *Helps protect body from infection*

What is the function of a lymphocyte?

- *Produces antibodies*

What is the function of a phagocyte?

- *Engulf pathogens (phagocytosis)*

Blood component	**Function**
Red blood cell	Transport oxygen bound to haemoglobin
White blood cell	Phagocytosis and antibody production
Platelet	Used in clotting
Plasma	Transports all blood cells and dissolved substances

Describe the process of clotting

- *Conversion of fibrinogen to fibrin*
- *Forms a mesh*
- *Prevents blood loss and entry of pathogens*

What happens at the capillary bed?

- *Artery brings oxygen and nutrients needed by tissue*
- *Substances move out from plasma, forming tissue fluid*
- *Substances collected from cells*
- *Blood leaving capillary bed deoxygenated, low in nutrients*

10. Diseases and immunity

What is a pathogen?
- Disease causing organism

What is a transmissible disease?
- Disease where the pathogen can be passed from one host to another

How can pathogens be transferred?
- Direct contact (e.g. blood or body fluids)
- Indirectly (e.g. contaminated surfaces, food, in the air)

Describe the body's defences to pathogens
- Mechanical barriers (e.g. skin and hairs)
- Chemical barriers (e.g stomach acid)
- White blood cells (e.g. phagocytes and antibody production which can be enhanced by vaccination)

How do antibodies work?
- *Bind to antigens*
- *All pathogens have their own antigens with specific shape*
- *Antibodies are specific to shape of antigen*
- *Cause direct destruction*
- *OR marks them for destruction by other cells*

What is active immunity?
- *Producing antibodies*
- *Gained after infection or vaccination*

How do vaccines work?
- *Harmless pathogen given which has antigens*
- *Antigens trigger immune response*
- *Lymphocytes produce antibodies*
- *Memory cells produced for long term immunity*
- *Needed to control disease spread*
- *More people immune, less pathogen created, less spread = herd immunity*

Method to reduce disease	Importance
Hygienic food preparation	Contaminated food can contain pathogens or toxins that cause food poisoning
Good personal hygiene	Reduce chances of contracting or transmitting disease
Waste disposal	Waste and rubbish is source of infection
Sewage treatment	Reduce risk from contaminated human waste

Explain passive immunity
- *Short term defence against pathogens*
- *Antibodies acquired from another individual*
- *Memory cells are NOT produced*
- *Important to breast fed infants - antibodies passed to infant in mother's milk, required as infants have undeveloped immune systems*

What causes type-1 diabetes?
- *Immune system targeting and destroying own body cells*

11. Gas exchange in humans

What are the features of a gas exchange surface?
- Large surface area
- Thin surface
- Good blood supply
- Good ventilation with air
- Moist

What is the function of tracheal cartilage?
- *Provides stiffness*
- *Prevents collapse during inhalation*

Describe the process of breathing in (inhalation)
- *External intercostal muscles contract*
- *Ribs move up and out*
- *Diaphragm contracts and flattens*
- *Volume inside thorax increases*
- *Pressure decreases*
- *Air is sucked into the lungs*

Describe the process of breathing out
- *Internal intercostal muscles contract*
- *Ribs move down and in*
- *Diaphragm relaxes and become dome shaped*
- *Volume inside thorax decreases*
- *Pressure increases*
- *Air is forced out of the lungs*

Component	**Inspired air (%)**	**Expired air (%)**	***Explanation***
Oxygen	21	18	*Oxygen moves from lungs into blood*
Carbon dioxide	0.04	3	*CO_2 has moves from blood in to lungs*
Water vapour	variable	Saturated	*Water evaporates from inside alveoli*
Nitrogen	*78*	*78*	*Not used by body*
Temperature	*variable*	*37 celsius*	*Body heat transferred to air*

How do you test for carbon dioxide?
- Bubble air through limewater indicator
- Limewater turns milky/cloudy in presence of carbon dioxide y

In what ways does breathing change during exercise?
- Increased rate – more breaths per min
- Increased tidal volume – more air per breath

How does exercise affect breathing rate and depth?
- *Exercise produces more carbon dioxide in muscles due to increase in respiration*
- *Carbon dioxide in blood increases*
- *Detected by brain*
- *Causes increased breathing rate and depth*

How is the gas exchange surface protected from pathogens and particles?

- *Mucus produced by goblet cells*
- *Traps particles and pathogens*
- *Cilia on surface beat to move mucus*
- *Mucus moved up and out of lung*

12. Respiration

How is energy used in the body?
- Muscle contraction
- Protein synthesis
- Cell division
- Active transport
- Growth
- Nerve impulses
- Keeping body temperature constant

Define aerobic respiration
- Oxygen used by cells to break down glucose to produce energy
- Word equation: glucose + oxygen —> carbon dioxide + water (+ energy)
- *Balanced chemical equation:* $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_20$

Define anaerobic respiration
- Glucose broken down to produce energy (without using oxygen)
- Produces less energy per molecule glucose
- Word equation (in muscles): Glucose —> lactic acid
- Word equation (in yeast): Glucose —> alcohol + carbon dioxide (in yeast)
    - *Balanced equation* $C_6H_{12}O_6 \longrightarrow 2C_2H_5OH + 2\ CO_2$

What is oxygen debt?
- *Build up of lactic acid in muscles and blood during exercise*

How is the oxygen debt removed?
- *Removed by aerobic respiration in liver*
- *Fast heart rate continues after exercise to take lactic acid from muscles to liver*
- *Deep breathing continues after exercise to continue supplying oxygen for aerobic respiration in liver*

13. Excretion in humans

How is carbon dioxide excreted?
- Through the lungs

What role does the liver play in excretion?
- Produces urea from excess amino acids
- *Liver assimilates amino acids in to proteins*

What is deamination?
- *Removal of nitrogen containing part of amino acids*
- *Forms urea*

What does the kidney excrete?
- Urea
- Excess water
- Salts

What affects the volume and concentration of urine?
- Water intake
- Temperature
- Exercise

Why is excretion needed?
- *Urea and carbon dioxide are toxic when allowed to build up in blood*

Function of glomerulus:
- *Filters blood of water, glucose, urea and salts*

Function of tubule:
- *Reabsorb all the glucose, most of water and some salts*
- *Leads to concentrated urea in urine*
- *Loss of excess water and salts*

What is dialysis?
- *Cleaning blood of urea, water and salts using a machine*
- *Arterial blood from body goes through machine*
- *Urea leaves blood through partially permeable membrane into dialysis fluid*
- *Composition of dialysis fluid means useful solutes not lost*
- *Blood returned to body in to vein*

	Dialysis	***Transplant***
Advantages	*No surgery needed*	*Cheaper in long run* *Less disruption*
Disadvantages	*Disruptive to life (3x/week)*	*Possible tissue rejection* *Surgery needed*

14. Coordination and response

Describe a nerve impulse
- Electrical signal that passes along neurones

What makes up the human nervous system?
- Brain
- Spinal cord
- Peripheral nervous system
- Coordination and regulation of body functions
- *Voluntary actions involve the brain, involuntary do not*

Describe a reflex arc
- Involuntary action
- Receptor receives stimulus
- Impulse travels down sensory neurone to spine
- Relay neurone conducts impulse across spine
- Motor neurone carries impulse to effector
- Effector carries out action

What is a reflex action?
- A way to automatically and rapidly respond
- Stimulus - receptor - sensory neurone - relay neurone - motor neurone - effector - response
- Involves electrical impulses and synapses
- E.g. Withdrawal of finger from hot object

What is a synapse?
- Junction between two neurones
- Neurotransmitter diffuses and binds to post-synaptic membrane

Describe a synapse
- *Endplate of one neurone and dendrite of next*
- *Neurotransmitter at end plate in vesicles*

How does an impulse cross a synapse?
- *Electrical impulse arrives at endplate*
- *Impulse causes neurotransmitter to transfer from vesicles to synaptic gap*
- *Chemical diffuses across gap and binds with receptor on next neurone*
- *Sets off impulse in next cell*
- *Synapse means that impulses travel in* ***one direction*** *only*
- *Some drugs (e.g. heroin) act at synapses to block binding of neurotransmitter*

What is a sense organ?
- Group of receptor cells that respond to specific stimuli
- Stimuli include: light, sound, touch, temperature, chemicals

Function of eye components
- Cornea – refracts light
- Iris - contains radial and circular muscles which control the size of the pupil
- Lens - refracts light to focus it on to retina
- Retina - contains photoreceptors (rods (dim light) and cones (detect colour)) which are sensitive to light
- Optic nerve - takes electrical impulses from the eye to the brain

How does the diameter of the pupil change in bright and dim light?
- Bright light – pupil diameter decreases to protect retina
- Dim light - pupil diameter increases to let in more light

How does the pupil constrict in bright light and why is this necessary?
- *Circular muscles contract*
- *Radial muscles relax*
- *Pupil constricts*
- *Protects the retina from the bright light*

How does the pupil dilate in dim light and why is this necessary?
- *Circular muscles relax*
- *Radial muscles contract*
- *Pupil dilate*
- *Allows more light to enter the eye*

What is accommodation?
- *Changes that take place within the eye*
- *Enable us to focus on objects at different distances*

How does the eye focus on a nearby object?
- *Ciliary muscle contracts*
- *Suspensory ligaments slacken*
- *Lens fat*
- *Light refracted strongly*

How does the eye focus on a faraway object?
- *Ciliary muscle relax*
- *Suspensory ligaments taut*
- *Lens thin*
- *Light refracted less strongly*

How are rods and cones distributed?
- *Rods around the edge and cones in centre of retina*

What do rods do?
- *See in black and white*
- *Pick up light at low levels*

What do cones do?
- *See in colour (red, blue, green)*
- *Give detail in bright conditions*

Where is the fovea?
- *Centre of the retina*

What is a hormone?
- A chemical substance produced by a gland
- Travels in blood
- Alters activity of target organ(s)

Hormone	Where produced	Function / Effect
Adrenaline	Adrenal gland	• Fight or flight response • Increased breathing rate and pulse, widened pupils, blood diverted from gut to muscles
Insulin	Pancreas	• Makes tissues absorb glucose from blood
Testosterone	Testes	• Stimulates secondary sexual characteristics e.g. voice deepening, sperm production, pubic hair
Oestrogen	Ovary	• Stimulates secondary sexual characteristics e.g. hips widening, breast growth, pubic hair • Repairs uterus lining • Inhibits FSH production, stimulates LH production

Examples of times adrenaline produced
- Running a race
- Chased by a predator
- Stressful situations

What is the role of adrenaline in a fight or flight situation?
- *Increases availability of glucose for muscles*
- *Increases pulse rate to deliver more blood to tissues*
- *Body ready for action - muscles primed to respond*

What is the difference between hormonal and nervous responses?
- *Nervous involves electrical impulses, hormonal involves chemicals carried in the blood*
- *Nervous response faster, hormonal slower*
- *Nervous response short-lived, hormonal long-lived*
- *Nervous response very localised, hormonal wide-spread*

Define homeostasis
- Maintenance of a constant internal environment
- *Control of internal conditions within set limits*
- *Controlled by negative feedback*

What is negative feedback?
- *The body's response when internal conditions deviate from the ideal set point*
- *Body returns conditions to this set point*

How is glucose level in the blood controlled?
- *Low glucose in blood detected —> glucagon released —> glycogen in liver released in to blood as glucose —> blood glucose increases*
- *Blood glucose detected too high —> pancreas secretes insulin —> tissues take up glucose —> blood glucose drops*

What are the symptoms and treatment of type-1 diabetes?

- *Symptoms:*
    - *Excessive thirst, urine production and hunger*
    - *Sweet smelling breath*
    - *Glucose in urine*
- *Treated by regular injections of insulin*

How is body temperature kept constant?

- Blood temperature receptors in brain detect change in blood temperature - coordinates response
- Sweating loses heat by evaporation
- Hairs trap insulating air beneath them to reduce heat loss
- Shivering produces heat from respiration and friction
- *Heat loss regulated by control of capillaries in skin*
- *Vasodilation allows more blood to reach skin surface - blood cools*
- *Heat conserved when skin capillaries vasoconstrict*

Define gravitropism

- Response of plant to grow towards/away from gravity

Define phototropism

- Response of a plant towards/away from light

Explain why phototropism and gravitropism are under chemical control

- *Chemical control of plant growth*
- *Auxin made in in shoot tip*
- *Auxin spreads through plant from shoot tip*
- *Auxin causes cells to lengthen*
- *Auxin gets unequally distributed due to light and gravity*

What is the role of the synthetic plant hormone 2,4-D in weedkillers?

- *Similar effect to auxins*
- *Selective - affects broad-leaves weeds but not grasses*
- *Increases rate of respiration and growth of weeds*
    - *Weeds exhaust their food supply and die*

15. Drugs

What is a drug?
- A substance taken in to the body
- Modifies or affects chemical reactions

What is an antibiotic?
- Drug used to treat bacterial infection
- Does not affect viruses
- Some bacteria are resistant to antibiotics which reduces their effectiveness

How can antibiotic resistance be minimised?
- *Using antibiotics only when essential*
- *Ensuring treatment is completed*

Why do antibiotics not affect viruses?
- *Antibiotics work by stopping bacterial replication and metabolism*
- *Viruses don't replicate the same way as bacteria*
- *Viruses live inside cells of the host*

What are the effects of excess alcohol or taking heroin?
- Powerful depressant drugs causing much slower reaction times
- Reduced self-control
- Cause addiction and withdrawal symptoms
- Negative social impact
- Injecting heroin can cause HIV infection
- Excessive alcohol can cause liver damage as alcohol broken down in liver

How does heroin affect the nervous system?
- *Acts on synapse and alters function*

Describe the effects of tobacco smoking
- Can cause COPD (chronic obstructive pulmonary disease)
- Can cause lung cancer
- Smoke affects gas exchange surfaces by heat
- Drying effect
- Cilia destroyed
- Can cause emphysema
- Harmful chemicals in smoke
- Carbon monoxide- reduces oxygen supply
- Nicotine – stimulant, causes addiction
- Tar – Irritant, causes cancer

Discuss evidence linking smoking and lung cancer
- *Sir Richard Doll's experiments*
- *Prospective study showing that early death more likely in smokers*
- *Retrospective study looking at lifestyle of people that died of lung cancer*
- *Carried out studies on doctors so that job wasn't a factor*
- *Separated groups living in country vs city*

Discuss the use of hormone use in sport
- *Can be used to improve performance*
- *Anabolic steroids increase muscle and reduce fat*
- *Testosterone stimulates male traits like aggression*

16. Reproduction

What is asexual reproduction?
- Producing genetically identical offspring
- One parent
- Examples: bacteria, propagated plants

	Advantage	*Disadvantage*
Sexual	• *Variation* • *New features allow adaptation to new environments* • *Crops- develop new varieties which grow better*	• *Two parents needed* • *Fertilisation is random* • *Can have harmful variants*
Asexual	• *One parent only* • *Faster- quick colonisation* • *Crops- produce large numbers of identical plants*	• *No variation* • *Won't adapt so problems will affect all individuals* • *Crops- cloned plants can all be susceptible to same disease*

What is sexual reproduction?
- Fusion of nuclei from two gametes (sex cells) to form zygote
- Produces genetically different offspring
- *Gamete nucleus is haploid*
- *Zygote nucleus is diploid*

What is fertilisation?
- Fusion of gamete nuclei

	Function
Sepal	Protect flower bud. Green
Petal	Attraction of insects. Bright colours May produce nectar
Anther	Male reproductive part Pollen grains contain male nucleus (gamete)
Stigma	Platform for pollen grains to land
Ovaries	Female reproductive part Contains female nucleus in an ovum

	Wind pollinated	**Insect pollinated**	**Explanation**
Stigma	Long, feathery, sticks out	Flat or lobed Where insects will brush	Feathery stigmas to catch pollen
Petals	Small, dull, no nectar	Large, bright, nectar	Insects attracted to bright colours
Anther	Hang loosely Thin filaments	Stiff Firmly attached	Pollen more easily dislodged by wind
Pollen	Large quantities, light, smooth	Small numbers Large, sticky	Sticky grains stick to insects

What is pollination?

- Transfer of pollen grains from anther to stigma

What is self-pollination?

- *Transfer of pollen grains from anther*
- *To stigma on flower of same plant*
- *Less variation*
- *Less able to respond to changes in environment*
- *More efficient*
- *Relies on pollinators*

What is cross-pollination?

- *Pollen transfer to different plant of same species*
- *More variation*
- *Can adapt to environmental changes*
- *Risky- depends on proximity/chance of pollination*

What is fertilisation?

- When pollen nucleus fuses with ovule nucleus
- *Pollen grain landing on stigma releases chemical signals*
- *Pollen tube grows down through style*
- *Acts as channel to deliver male gamete to ovule*
- *Tip of pollen tube locates micropyle on ovule*
- *Male gamete enters ovule through micropyle*

What do seeds need for germination?

- Suitable temperature
- Oxygen
- Water

What is fertilisation (in a human)?

- Fusion of nuclei from a male gamete (sperm)
- With a female gamete (ovum/egg cell)

Compare male and female gametes:

- *Sperm are very mobile, numerous, small*
- *Ova (egg cells) are immobile, singular, large*

What are the adaptive features of sperm?

- Flagellum *for movement*
- Enzymes *to gain entry to ovum*
- *Mitochondria to supply energy for beating of flagellum*
- *Acrosome enzymes to penetrate egg cell membrane*

What are the adaptive features of egg cells (ova)?

- Energy stores
- Jelly coat that changes at fertilisation
- *Energy stores to supply energy for early development*
- *Jelly coat changes at fertilisation to allow entry of male nucleus*

What happens after fertilisation?

- Zygote forms embryo
- Embryo is ball of cells that implants into uterus wall

	Function	Description
Umbilical cord	Blood supply Connects fetus to placenta	*Carries materials for exchange between mother and fetus*
Placenta	Supply nutrients/ exchange waste Physical attachment Protects from blood pressure changes and mother's immune system Secretes hormones	*Exchanges soluble materials e.g. nutrients, wastes and oxygen.* *Provides a barrier to toxins and pathogens* *Some toxins can e.g. nicotine can pass across placenta*
Amniotic sac	Encloses amniotic fluid	
Amniotic fluid	Fluid surrounding fetus Protects from mechanical shock	

Describe the growth and development of a fetus
- Complexity increases in early stages
- Size increases towards the end of pregnancy

Why should women avoid alcohol in pregnancy?
- Can cross to fetus and cause slow development
- Causes fetal alcohol syndrome

Why should pregnant women avoid smoking?
- Reduces oxygen delivery to fetus and nicotine crosses placenta

	Advantages	*Disadvantages*
Breast feeding	*Ideal nutrient content* *Antimicrobial factors* *Low cost* *Encourages bonding* *No preparation needed*	*Others cannot feed* *Can't measure intake*
Bottle feeding formula	*Exact quantity measurable* *Other people can help feed*	*More expensive* *Not as easily digested* *Can pass microbes to baby*

Describe labour and birth
- Breaking of amniotic sac
- Muscles in uterus wall contract and cervix dilates
- Baby passes through vagina
- Cord is tied and cut
- Afterbirth delivered

What is the role of testosterone?
- Behaviour changes, aggression, territorial, attracted to girls
- Facial and chest hair
- Broader chest, larger muscles, deeper voice
- Testes produce sperm
- Penis and scrotum get larger

What is the role of oestrogen?

- Behaviour change- more maternal, attracted to boys
- Cycle of egg production starts
- Breasts increase in size and vagina gets larger
- Hair under armpits and pubic region
- Hips get broader, pelvis wider

Where are oestrogen and progesterone produced?

- *Ovary produces both during menstrual cycle*
- *Progesterone secreted by corpus luteum after implantation*
- *Uterus produces progesterone during pregnancy*

Explain hormone control of menstruation

- *LH stimulates release of mature ovum*
- *LH stimulates corpus luteum*
- *Progesterone keeps uterus ready for implantation*
- *Oestrogen repairs uterus lining*
- *FSH stimulates development of Graafian follicle*
- *Progesterone maintains uterus lining in pregnancy*

Methods of birth control

Natural	Chemical	Barrier	Surgical
Abstinence	IUD	Condom	Vasectomy
Body temperature monitoring	IUS	Femidom	Female sterilisation
Cervical mucus	Contraceptive pill	Diaphragm	
	Implant		
	Injection		

How are hormones used in contraception

- *Progesterone only makes implantation difficult*
- *Combined pill (oestrogen and progesterone) stops ovulation*

What is artificial insemination

- *Sperm from a donor*
- *Kept at sperm bank*
- *Inserted to woman's vagina near ovulation*

In Vitro Fertilisation (IVF)

- *Ovum fertilised outside the body*
- *In a laboratory*
- *Fertilised ovum placed in to woman's uterus*
- *Helps couples struggling to conceive*

What is a sexually transmitted infection?

- Infection transmitted by bodily fluids through sex
- E.g HIV

How is spread of STIs controlled?

- Individuals can use condoms, know partners sexual history and have medical checks
- Communities can offer testing
- Tracing sexual contacts to identify source
- Worldwide education programmes
- Providing antibiotics, vaccines and antiviral drugs

How is HIV transmitted?

- Unprotected sex with an infected person
- Contact with infected blood
- Mother to child
- Sharing syringes

How does HIV affect the body?

- Can lead to AIDS
- *Affects immune system - reduces lymphocyte numbers*
- *Reduced ability to produce antibodies*

17. Inheritance

What is inheritance?
- Transmission of genetic information down generations

What is a chromosome?
- Thread like structure of DNA
- Carries genetic information in the form of genes

What is a gene?
- Length of DNA that codes for a protein

What is an allele?
- Different form of the same gene which gives rise to different characteristics

How is sex inherited in humans?
- One sex chromosome inherited from each parent
- Inherit an X from the mother, and either X or Y from father
- Females are XX
- Males are XY
- 50:50 chance of being male or female

Describe base pairing
- *Adenine (A) - Thymine (T)*
- *Cytosine (C) - Guanine (G)*

Describe the structure of DNA
- *2 strands*
- *Coiled to form double helix*
- *Strands linked by paired bases*
  - o *Adenine to thymine*
  - o *Guanine to cytosine*

Describe the structure of RNA
- *Single stranded*
- *Contains uracil (U) base instead of thymine (T)*

How does DNA code for proteins?
- *Each DNA nucleotide codes for an RNA nucleotide*
- *Three nucleotides code for one amino acid*
- *Sequence of bases in a gene is code for sequence of amino acids in a protein*
- *Specific sequence= specific protein*

How does DNA control cell function?
- *Controls production of proteins*
- *Proteins include enzymes, antibodies and receptors for neurotransmitters*

How is a protein made?
- *Gene coding for protein stays in nucleus*
- *mRNA carries a copy to the cytoplasm*
- *mRNA passes through ribosomes*
- *ribosome assembles amino acids in to proteins*
- *Specific order of amino acids determined by sequence of bases in mRNA*

Are all proteins made in all cells?

- *No*
- *All cells contain all genes but many genes not expressed*
- *Each cell only makes the proteins it needs*

What is a haploid nucleus?

- *Nucleus containing single set of unpaired chromosomes e.g. gametes*

What is a diploid nucleus?

- *Nucleus containing two sets of chromosomes e.g. in body cells*
- *One pair of each type of chromosome*
- *23 pairs in a human*

What is mitosis?

- Nuclear division producing genetically identical cells
- Needed in growth, repair and replacement of cells
- Cell division for asexual replication

How does mitosis happen?

- *Chromosomes duplicate first*
- *Copies of chromosomes separate*
- *Chromosome number is maintained*

What is a stem cell?

- *Has the potential to divide many times whilst remaining undifferentiated (unspecialised)*
- *Divides by mitosis to produce daughter cells*
- *Daughter cells can become specialised for specific functions*

What is meiosis?

- Nuclear division producing genetically different cells
- Involved in producing gametes
- *Reduction division - chromosome number is halved (from diploid to haploid)*
- *Produces variation*
    - o *New combinations of maternal and paternal genes made*

What is genotype?

- Genetic make up of an organism i.e. which alleles are present

What is phenotype?

- Observable features of the organism

Define homozygous

- Having two identical alleles of a gene
- Two homozygous parents will produce pure-bred offspring

Define heterozygous

- Having two different alleles of a gene
- A heterozygous individual cannot produce pure-bred offspring

Define dominant

- An allele that is expressed where it is present

Define recessive

- Allele only expressed when there is no dominant allele present

Phenotypic ratios

- A_bA_B mother and A_bA_B father
- Possible offspring (punnet square)

		Father	
		A_b	A_B
Mother	A_b	A_bA_b (recessive)	A_bA_B (dominant)
	A_B	A_BA_b (dominant)	A_BA_B (dominant)

- Outcomes 3:1 dominant to recessive

What is a test-cross used to show?

- *To identify an unknown genotype*

A test-cross:

- *Phenotype- dominant characteristic. Use the recessive phenotype to test*
- *Genotype = Could be Bb or BB*
- *Gametes = B b x b b OR B B x b b*
- *Fertilisation = 50% dominant to 50% recessive OR all dominant*
- *So if any offspring have the recessive phenotype, the parent was heterozygous*

What is co-dominance?

- *More than two alleles occur*
- *Some alleles are not dominant to each other*
- *An extra phenotype will exist e.g. ABO blood groups (see table)*

Genotype	***Phenotype (blood group)***
I^AI^A Or I^AI^O	*A*
I^BI^B OR I^BI^O	*B*
I^AI^B	*AB (co-dominant)*
I^OI^O	*O*

What is a sex-linked characteristic?

- *Gene responsible located on sex chromosome*
- *More likely to occur in one sex than the other - example is colour blindness*

Sex linkage genetic diagram for colour blindness:

- *Let C = normal allele*
- *c = colour blind allele*
- *Carrier woman x normal man —> X^CX^c x X^CY*
- *Gametes = X^C X^c X^C Y*
- *Outcomes =*

	X^C	X^c
X^C	X^C X^C	X^C X^c *(carrier female)*
Y	X^C *Y*	X^c *Y (colour blind male)*

18. Variation and selection

What is variation?
- Differences between individuals of same species

What is phenotypic variation?
- Different observed characteristics
- May have different genes
- *Caused by genetic and environmental factors*

What is genetic variation?
- Different genes between individuals
- May have the same phenotype

What is continuous variation?
- Range of phenotypes between two extremes e.g. height in humans

What is discontinuous variation?
- Limited number of phenotypes with no intermediates - e.g. tongue rolling
- *Caused by genes alone e.g. A, B, AB, O blood groups*

What is mutation?
- Genetic change
- *Altered base sequence of DNA*

How are new alleles made?
- Through mutation

How is rate of mutation increased?
- By ionising radiation
- Certain chemicals

What are the symptoms of sickle cell anaemia?
- *Homozygous sufferers very anaemic, die without medical care*
- *Heterozygotes some symptoms, mild anaemia, may be weak*
- *Carrying the recessive allele gives resistance to malaria*

How does sickle cell occur?
- *Change in base sequence haemoglobin gene*
- *Abnormal haemoglobin*
- *Sickle shaped red blood cells*

Where is sickle cell anaemia most common?
- *In regions with high malaria*

Define an adaptive feature
- Inherited feature helping an organism to survive and reproduce
- *Inherited functional feature of an organism that increase its fitness*

Define fitness
- *Probability of an organism surviving and reproducing in their environment*

	Hydrophyte e.g. water lily	*Xerophyte e.g. cactus*
Adaptive features	*Little lignin* • *Supported by water* *Very thin cuticle* • *Water not a problem* *Stomata on upper surface* • *Uptake CO_2 from atmosphere*	*Green stem* • *Carries out photosynthesis* *Leaves reduced to spines* • *Reduce surface area for water loss* *Swollen stem* • *Stores water* *Deep roots*

What is natural selection?
- Variation within populations
- Many offspring produced
- Competition for resources
- Struggle for survival
- Individuals that succeed reproduce
- Passing their alleles to next generation

Define evolution
- *Change in adaptive features of a population over time*
- *Result of natural selection*
- *An example is bacterial resistance to antibiotics*

What is the process of adaptation?
- *Process resulting from natural selection*
- *Populations become more suited to their environments*
- *Over many generations*

What is selective breeding?
- Humans select individuals with desirable features
- These are crossed to produce next generation
- Offspring show desirable traits

Artificial selection vs Natural selection
- *Humans replace the environment in artificial selection*
- *Humans select individuals with most useful characteristics*
- *Humans allow only these to breed*
- *Examples are dairy cattle and domestic dog breeds*
- *Used to improve crops*
- *Carried out over many generations*

19. Organisms and their environment

What is the main energy source for biological systems?
- The sun
- *Light energy transferred to chemical energy in organisms*
- *Energy flow through living organisms*
- *Transferred to the environment*

Define a food chain
- Shows transfer of energy from one organism to the next
- Begins with a producer

How is energy transferred in a food chain?
- Energy transferred by ingestion
- Producers make their own nutrients using sun's energy
- Consumers feed on other organisms
- Primary, secondary and tertiary consumers classed by position in food chain
- *Energy is transferred between trophic levels*
- *Sun's energy fixed by producers*
- *Digestible plant matter consumed to give energy – 5-10% efficient*
- *Animal matter consumed as food gives 10-20% efficiency*

Define trophic level
- *Position of an organism in a food chain, web, pyramid of numbers or biomass*

What are the trophic levels?
- *Producers*
- *Primary consumers*
- *Secondary consumers*
- *Tertiary consumers*
- *Quaternary consumers*

Why is energy transfer inefficient?
- Energy is lost at every step
- Respiration transfers energy to environment
- Energy lost as heat

Why are food chains usually less than 5 trophic levels?
- Too little energy left for upper levels to survive

Why are plants more efficient for human food?
- *Plants are 1st trophic level*
- *Less energy lost by eating plants*
- *80-90% of energy lost in feeding plants to animals then eating the animals*

What is a food web?
- Network of interconnecting food chains

What is an herbivore?
- Animal that gets its energy from eating plants

What is a carnivore?
- Animal that gets energy by eating animals

What is a decomposer?
- Organism that gets energy from dead or waste organic material

What impact have humans had on food chains?
- Over harvesting of food species deprives consumers of food
- Consumers may switch to eating other foods
- Introducing foreign species means more competition for food
- Less food available for others

What is a pyramid of numbers?
- Diagram representing numbers of organisms at each trophic level
- In a given ecosystem at any one time

What is a pyramid of biomass?
- *Represents the biomass (number of individuals x their mass)*
- *At each trophic level at any one time*

What are the advantages of a pyramid of biomass?
- *Solves scale and inversion problems with pyramid of numbers*

Describe the carbon cycle
- CO_2 converted to organic compounds during photosynthesis using light energy
- Respiration makes CO_2 from organic compounds to produce energy
- Feeding transfers organic compounds in plants to animals
- Decomposers release CO_2
- Combustion releases CO_2 by burning of fossil fuels
- Fossilisation occurs where conditions prevent decomposers
- Fossilisation creates fossil fuels, this lowers the CO_2 concentration in the environment

What is the effect of burning fossil fuels and cutting down forests?
- CO_2 in fossil fuels and trees is released in to the atmosphere
- Levels of CO_2 increase

Describe the water cycle
- Water evaporates from the earth's surface turning to vapour
- Vapour condenses in the atmosphere creating clouds
- Cloud vapour turns to droplets, falling as rain, hail and snow (precipitation)
- Water on earth's surface freezes solid in to ice
- Ice melts back in to water

Describe the nitrogen cycle
- *Nitrogen in atmosphere as N_2 is fixed by lightning and bacteria*
- *Nitrogen fixation makes nitrate ions in soil, absorbed by plants*
- *Plants make organic compounds*
- *Animals feed on and digest organic compounds in plants*
- *Animals produce waste products by deamination of proteins*
- *Decomposition of plant and animal protein makes ammonium ions*
- *Ammonium ions converted to nitrate ions by nitrification*
- *Nitrate ions converted to Nitrogen gas by denitrification*

What are the roles of microorganisms in the nitrogen cycle?
- *Decomposing bacteria make ammonium ions*
- *Nitrogen fixing ions make nitrate ions from N_2*
- *Nitrifying bacteria make nitrate from ammonium ions*
- *Denitrifying bacteria make nitrogen gas from nitrate*

Define population
- Group of organisms of one species living in same area at same time

Define community
- *Populations of different species in an ecosystem*

Define ecosystem

- *Unit containing community of organisms and their environment interacting together*

What factors affect the rate of population growth for an organism?

- Biotic factors:
    - Food supply – enough to support population
    - Predation- more predation reduces numbers
    - Disease – reduces numbers
- Abiotic factors:
    - Temperature
    - Oxygen
    - Light
    - Toxins and pollutants

Discuss human population increase over the past 250 years

- Agricultural revolution - better able to create food
- Industrial revolution - agriculture more efficient, people earn more money
- Medical revolution – better health care, people live longer
- Environment could support more people, people live longer
- Massive increases in population
- Caused greater competition for land
- Greater effects on environment
- More demand for leisure space, housing, food and roads

Describe and explain the shape and phases of a population growth curve

- *Sigmoid shape*
- *Lag phase: Population growth begins slowly from a few individuals*
- *Log phase: Exponential growth occurs. Conditions are ideal and maximum growth rate is reached*
- *S-phase: Growth rate begins to slow down. Factors such as food, water and space become limiting*
- *Stable phase: Carrying capacity (the population size that can be supported by a particular environment) reached. Population number becomes stable*
- *Decline phase: If sudden change in the environment (e.g. a drought causing food shortage), population will crash and the whole process begins again*

20. Biotechnology and genetic engineering

Why are bacteria useful in biotechnology?
- Rapid reproduction rate
- Ability to make complex molecules
- *No ethical concerns over manipulation and growth*
- *Genetic code shared with all other organisms*
- *Plasmids present*

How is yeast used to make biofuel?
- Used to produce ethanol
- Anaerobic respiration in yeast turns sugar and other nutrients to alcohol
- Then distilled to make pure ethanol

How is yeast used in bread making?
- Yeast mixed in to bread mixture
- In warm environment yeast ferments sugar
- Anaerobic respiration produces CO_2 bubbles
- Bubbles make dough rise

How is pectinase used in fruit juices?
- Pectinase breaks down clumps of plant cells
- Add to fruit juice - Juice goes from cloudy to clear

Describe enzyme use in washing powders
- Biological washing powder contains lipase and proease
- Stains made of lipids and proteins
- Broken down by enzymes. Doesn't affect material

How is lactase used to produce lactose free milk?
- *Enzymes immobilized on fibres*
- *Milk passed over immobilized lactase enzyme*
- *Enzyme breaks down lactose*
- *Product passes out lactose free*

How is fungus used to make penicillin?
- *Penicillum fungus produces antibiotic to kill off other microorganisms*
- *Made when growth of fungus is slowing down*
- *Penicillin made in industrial fermenters to grow fungus and harvest antibiotic*
- *Made by batch fermentation - Batch collected and fungus filtered out*
- *Filtered liquid contains antibiotic*

Define genetic engineering
- Changing an organism's genetic material by removing, adding or changing genes

Give examples of genetic engineering
- Inserting human genes in bacteria to make insulin
- Adding genes to crop plants to give herbicide resistance
- Adding genes to crops to give pest resistance
- Adding genes to crops to give extra vitamins

How does genetic engineering work?
- *Restriction enzymes isolate human DNA, leaving sticky ends*
- *Bacterial plasmid cut with restriction enzymes, matching sticky ends*
- *Human DNA inserted in to bacterial plasmid using DNA ligase*
- *Recombinant plasmid made and inserted in bacteria*
- *Bacteria replicated, making human protein when genes expressed*

Human proteins commonly made

- *Insulin, growth hormone, factor VIII (haemophilia)*

Advantages and disadvantages of genetically modifying crops

Advantages	*Disadvantages*
Higher yields with fewer resources e.g. pesticide, fertiliser	*Cross pollination with weeds can make superweeds*
Crops grow in more extreme environments, more crops grown, less famine	*Engineered bacteria could escape- unpredictable consequences*
More predictable than selective breeding	*New organisms could be patented, companies may become very powerful*
Foods can be made more convenient	*Pushing boundaries- moral/ethical question*

21. Human influences on ecosystems

How has modern technology increased food production?

- Agricultural machinery covers more land, more efficient
- Chemical fertiliser improves yields
- Insecticides improve quality and yield
- Herbicides reduce competition
- Selective breeding improves production

Describe negative impacts to ecosystems of large monoculture

- Poor wildlife food
- Disease can spread from crop to wild
- Loss of genetic variety
- Can damage soil by mineral loss

Describe negative impacts to an ecosystem of intensive livestock production

- Deforestation used to clear land- loss of habitat
- Removal of hedges – shelter, feeding, breeding sites for wildlife
- Deforestation reduces soil fertility
- Can lead to landslips, flooding

Discuss social, environmental and economic implications of providing food for the global population

- *Food unevenly distributed - Food mountains in rich countries*
- *Some countries feel poor people should become self sufficient and reserve food aid for disasters*
- *Transporting food from rich to poor countries not economic*
- *Perishable foods may not be able to travel*
- *Cheap food from rich countries can upset local economy*
- *Low crop yield in poor countries*
- *Malnourished people get ill more easily, may have to care for relatives so cannot work*

Problems contributing to famine

- *Unequal distribution of food*
- *Drought or flooding*
- *Increasing population*
- *Poverty*

Describe the reasons for habitat destruction

- Increased area for food crop growth, livestock and housing
- Extraction of natural resources
- Marine pollution

What are the negative effects of deforestation?

- Habitat destruction
- Extinction
- Soil loss
- Flooding
- Increased CO_2 in atmosphere
- Climatic changes

Explain the effects of deforestation

- *Trees contain up to 90% of the nutrients in a forest ecosystem - Taken away if trees removed*
- *Many species depend on trees, lose habitat and die*
- *Loss of soil structure from roots contributes to erosion*
- *No trees to absorb rainfall, causing flooding and landslips*
- *Fewer trees produce less CO_2 and absorb less O_2*
- *Reduced transpiration and drier atmosphere affects water cycle*
- *Bare soil absorbs heat and increases winds*

	Sources	**Effects**
Insecticides	Water treatment for aquatic insects Crop spraying	High concentrations in top predators Can be toxic or harmful
Nuclear fall out	Nuclear power plant accidents Use of nuclear weapons	Makes land uninhabitable Contaminated marine life Death from radiation exposure
Chemical waste	Oil spills	Reduce oxygen on seabed Harm seabirds
Discarded rubbish	Human littering	Block water passage Eaten by animals
Untreated sewage	Human sewage systems Slurry run off	Depleted oxygen- death of fish and invertebrates
Fertilisers	Farming practices	Depleted oxygen – death of fish and invertebrates
Methane	Farm animals Waterlogged swamps, rice field	Global warming Climate extremes, ice caps melt
Carbon Dioxide	Fossil fuel combustion	Global warming More deserts Pests can spread

Describe eutrophication

- *Raw sewage and leaching of inorganic fertilisers - More nitrate and phosphate*
- *Increased growth of producers*
- *More decomposition when producers die so decomposition increases*
- *More anaerobic respiration from decomposers —> Less dissolved oxygen*
- *Organisms that need oxygen die*

Discuss impact of non-biodegradable plastics

- *Blocked passage of water —> waterlogging*
- *Soils less oxygenated, less fertile*
- *Animals consume plastics by mistake causing death*
- *Stop natural decomposition of other waste when in landfill*
- *Release toxic smokes when burned*

Discuss causes and effects of acid rain
- *Sulphur and nitrogen oxides made by combustion are further oxidised in clouds*
- *Dissolve to form acids that fall as acid rain*
- *Soils become acidic - Leaches minerals and inhibits decomposition*
- *Water in lakes and rivers collects excess minerals and fish die*
- *Starvation of forest trees from damaged leaves and less minerals*

What are solutions to acid rain?
- *Clean up emissions from power stations with scrubbers*
- *Catalytic converters in cars*
- *Adding crushed limestone to lakes and fields*

What is the greenhouse effect?
- *Increases in CO_2 and methane enhances heat trapping in atmosphere*
- *More of suns energy reflected back to earth and leads to climate change*

What is the impact of female contraceptive hormones in watercourses?
- *Reduced sperm counts in men*
- *Feminisation of aquatic organisms e.g. Imbalanced gender in fish populations*

Define sustainable resource
- Produced as rapidly as removed
- Does not run out

Why do we need to conserve fossil fuels?
- They are a non renewable resource - will run out eventually

Which resources can be maintained?
- Forests and Fish stocks

Name resources that can be re-used or recycled
- Paper, Glass, Plastic, Metal

How is sewage treated to make water safe?
- Flow of water carries sewage away
- Taken to treatment plant
- Water can be recycled

Why do organisms become endangered or extinct?
- Climate change alters habitat
- Habitat destruction e.g. from deforestation
- Hunting
- Pollution
- Introduced species (competition for food/resources)

How can endangered species be conserved?
- Monitoring species and habitats
- Providing protection
- Education
- Captive breeding programmes
- Seed banks for plants

Define sustainable development
- *Providing for needs of increasing human population*
- *Without harming the environment*

How can fish stocks and forests be sustained?

- *Education*
- *Legal quotas prevent over fishing or deforesting*
- *Restocking*

What are the risks to a species if population drops?

- *Fewer individuals*
- *Have less variation*
- *Less genetic variation can lead to inbreeding*
- *Less able to breed and produce healthy offspring*

What are the reasons for conservation programmes?

- *Reducing extinction*
- *Protecting vulnerable environments*
- *Maintaining ecosystem functions*
- *Nutrient cycling*
- *Resource provision (food, drugs, fuel and genes)*